

Retrieval Perturbation as a Stress-Test for RAG Faithfulness: A Counterfactual Benchmark for Evaluating Graceful Degradation in Retrieval-Augmented Generation

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Abstract

Retrieval-augmented generation (RAG) depends on retrieved context, but common evaluations confound retrieval error with generator behavior. This paper introduces a counterfactual benchmark that changes a retrieved passage while holding the question, answer, and prompt fixed. The benchmark applies entity substitution, factual negation, and meaning-preserving paraphrase to 1,000 SQuAD v1.1 question-answer pairs. Six language models, four open-weight and two closed-source, are evaluated in every condition ($N = 24,000$ model-condition inputs). Relative to original context, mean NLI faithfulness on corrupted context decreases by 0.153–0.295 across models, and abstention rises by 13.4–30.9 percentage points; paired 95% confidence intervals exclude zero for every primary comparison. Hallucination is observed in 0–9% of outputs, with some zero-event cells having nonzero uncertainty. The dominant failure is therefore neither uniformly faithful following nor pervasive fabrication, but insufficiently selective abstention. The two closed-source models have the largest abstention-cliff scores (0.241 and 0.244), while the two Llama models have the smallest (0.173 and 0.192). A matched, 200-example similarity-score ablation lowers a descriptive abstention-calibration discrepancy but also raises abstention on clean context, so it does not improve selective abstention. Human validation of 80 perturbations identifies predictable weaknesses in rule-based negation and entity substitution. The benchmark isolates a concrete retrieval-generation failure mode and provides evidence that calibration depends more on model family than parameter count in this setting.

1 Introduction

Retrieval-Augmented Generation (RAG) has become the dominant architecture for grounding large language model (LLM) outputs in external knowledge [1]. By coupling a dense retriever with a generative model, RAG systems reduce hallucination and enable up-to-date answers without re-training. They are now widely deployed in enterprise search, legal document analysis, medical question answering, and customer-facing AI assistants. Retrieval is inherently noisy: retrievers regularly surface stale documents, near-duplicate passages with incorrect dates or entities, or topically adjacent but factually mismatched chunks. Retrieval failure is not an edge case but a routine feature of production RAG systems.

Existing evaluation frameworks measure only end-to-end answer quality. RAGAS [2] and ARES [3] score faithfulness and answer relevance, but both assume the retrieved context is the best available context and treat errors as a monolithic signal. When a RAG system gives a wrong answer, neither framework can determine whether the retriever surfaced the wrong document or

whether the generator failed to reason correctly from a good one. TruLens and the TREC RAG Track [4] similarly measure pipeline outputs without isolating fault.

That attribution gap prevents system designers from determining whether retrieval or generation warrants intervention. It also leaves model developers without a benchmark for the retrieval-generation interface under controlled failure conditions.

The benchmark addresses this gap by systematically modifying retrieved context in three ways and measuring the resulting generator response. Because the modification is known, the source of a context error is controlled. Four context conditions are defined for each question-answer pair:

- Original: unmodified gold context (baseline)
- Paraphrase: surface rewording with facts preserved (control)
- Entity swap: one named entity replaced with a plausible same-type alternative
- Negation: core factual claim directly contradicted via dependency-parse negation

The paraphrase condition serves as a control: a robust model should perform identically on paraphrase and original. Meaningful divergence on entity swap and negation reveals failure modes of practical interest.

This paper addresses four research questions:

- RQ1. Do LLMs faithfully follow corrupted context, even when it is wrong?
- RQ2. Do LLMs hallucinate when retrieved context conflicts with ground truth?
- RQ3. Do LLMs appropriately abstain (“I don’t know”) under corrupted retrieval?
- RQ4. Does model scale, family, or open- vs. closed-source status predict better graceful degradation?

The contributions are:

- A controlled counterfactual RAG benchmark built on SQuAD v1.1 [16], with 1,000 paired examples and four context conditions per model.
- An evaluation of six models spanning open-weight and closed-source families, with paired confidence intervals and corrected paired tests for the primary comparisons.
- A failure-mode taxonomy distinguishing three generator archetypes: context-follower, hallucinator, and over-abstainer, with evidence that model family is more predictive of abstention calibration than parameter count in the evaluated set.
- A matched similarity-score ablation and human validation of 80 perturbations, which qualify the scope of the empirical findings.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the benchmark design. Section 4 details the evaluation pipeline. Section 5 presents results. Section 6 provides analysis and the failure-mode taxonomy. Section 7 discusses implications. Section 8 states limitations. Section 9 concludes.

2 Related Work

RAG evaluation. RAGAS [2] and ARES [3] score faithfulness and answer relevance using LLM judges, and the TREC RAG 2024 Track [4] provides large-scale information-retrieval-style evaluation. These frameworks evaluate the final answer and retrieved context together, without holding retrieval quality constant. The present benchmark instead varies only the content of retrieved context, supporting attribution of behavioral differences to the generator.

Robustness to adversarial context. Prior work on adversarial reading comprehension includes AddSent [5], which appends distracting sentences to SQuAD passages, contrast sets via minimal manual perturbations [6], and counterfactual data augmentation [7]. The present benchmark targets a different problem: it simulates plausible wrong entities and contradicting documents, then measures abstention calibration, a behavior not emphasized by prior robustness benchmarks.

Hallucination, abstention, and irrelevant context. TruthfulQA [8] measures false statements on misconception-prone questions; calibration literature [9, 10] studies whether model confidence correlates with accuracy; and selective prediction [11] formalizes abstaining when confidence is low. Most relevant to this setting, Shi et al. [12] show that irrelevant context can mislead LLMs even when correct context is also present. This benchmark treats abstention as a first-class metric under a controlled perturbation protocol and connects abstention rates to specific retrieval failure modes. CheckList [13] and ANLI [14] establish the broader paradigm of behavioral testing via perturbations; this benchmark applies similar logic but centers on abstention calibration and production RAG failure modes.

Open-weight vs. closed-source model behavior. Recent work compares open-weight and closed-source LLMs on reliability and calibration tasks [9, 10]. The mechanisms underlying observed differences, including instruction tuning, preference optimization, and training-data composition, remain uncertain. By holding the prompt, task, and evaluation protocol constant, the present study measures abstention calibration as one axis of difference between the evaluated model families.

3 Benchmark Design

3.1 Seed Dataset

Seed data come from SQuAD v1.1 [16], a reading-comprehension dataset of Wikipedia passages paired with crowd-sourced questions and extractive answers. The following filters obtain clean, unambiguous examples:

- The gold context must be extractable as a single sentence containing the gold answer (the sentence is found by splitting on periods and selecting the first that includes the answer string verbatim; sentences shorter than 20 characters are skipped).
- The gold answer string must appear verbatim in the selected context sentence.

Filtering yields 1,000 question-answer pairs, sampled randomly from the full SQuAD v1.1 training split. Each example has the format:

$$\{\text{id, question, gold_answer, gold_context}\}$$

Table 1 reports dataset statistics.

Answer-type labels (person, organization, location, date/number) were derived during error analysis using named-entity recognition applied to gold answers. They are reported in Section 6.3 and were not used to construct the base dataset.

3.2 Perturbation Types

Four context variants are generated for each question-answer pair. Table 2 summarizes the conditions.

Table 1: Dataset statistics for the 1,000-example evaluation split.

Property	Value
Total QA pairs	1,000
Source	SQuAD v1.1 (Wikipedia)
Avg. gold context length (tokens)	27.2
Avg. question length (tokens)	10.0

Table 2: Perturbation conditions. Each QA pair is evaluated under all four conditions.

Condition	Transformation	Facts	Wording	Role
Original	None	✓	✓	Baseline / upper bound
Paraphrase	LLM surface rewrite	✓	×	Control: surface sensitivity
Entity swap	Same-type NER replacement	×	✓	Simulates wrong-entity retrieval
Negation	Dependency-parse negation	×	✓	Simulates contradicting retrieval

Original. The unmodified gold context sentence. It sets the upper bound for correctness and the expected lower bound for abstention: a well-calibrated model should rarely decline to answer when the context is correct and directly relevant.

Paraphrase. A language model rewrites the gold context sentence while preserving factual content. A robust model should produce similar values for paraphrase and original context. Divergence indicates sensitivity to surface wording rather than semantic content.

Entity swap. Named entities in the context are identified, and one entity with a supported type is replaced by a plausible alternative of the same type. For example, a person is replaced by another person and a location by another location. This condition simulates a retrieved document that retains the claim structure but contains a wrong key fact.

Negation. A rule-based transformation negates the central predicate of the gold context sentence while preserving the remaining content. This simulates a retrieved record or policy that directly contradicts the original claim.

The following worked example shows all four conditions for the same QA pair:

Question: Who co-founded Microsoft?

Gold answer: Bill Gates

Original: Microsoft was founded by Bill Gates and Paul Allen in 1975.

Paraphrase: Bill Gates and Paul Allen established Microsoft in 1975.

Entity swap: Microsoft was founded by Steve Jobs and Paul Allen in 1975.
Negation: Microsoft was not founded by Bill Gates and Paul Allen in 1975.

3.3 Perturbation Quality

Automatic transformations can introduce fluency errors or incoherent outputs, especially when negating complex sentences. The negation procedure handles auxiliary and simple-verb constructions, with a conservative fallback for other sentences. Entity-swap examples are created only when a named-entity type has a plausible replacement.

To validate pipeline quality, two independent annotators rated 80 randomly sampled examples (20 per condition) on two dimensions: fluency (1 = grammatically broken, 2 = slightly awkward, 3 = fluent and natural) and correctness (1 = perturbation failed, 2 = partial success, 3 = clear success). Inter-rater agreement was Cohen’s $\kappa = 0.396$ for fluency and $\kappa = 0.479$ for correctness. Mean fluency across all conditions was 2.725 and mean correctness was 2.475. Entity-swap errors concentrated in chemical compounds and domain-specific names. Paraphrase errors primarily altered quoted speech. Negation errors were most common in multi-clause sentences, where negating a central predicate could create a contradiction with a dependent clause. Section 8 discusses the implications of these quality limits.

4 Experimental Setup

4.1 Retriever

For each question-answer pair and context condition, a dense retriever supplies one passage ($k = 1$), using normalized Sentence-BERT embeddings [15]. The retrieved passage is the corresponding perturbed sentence. Consequently, every model receives exactly the intended context, and differences across conditions reflect response to content changes rather than variation in candidate retrieval.

4.2 Generator

Six models spanning open-weight and closed-source families are evaluated at temperature zero with the same prompt template.

- llama3.2:3b (3B parameters): smallest open-weight model
- llama3.1:8b (8B parameters): same family at larger scale
- mistral:7b (7B parameters): different instruction-tuning lineage
- qwen2.5:7b (7B parameters): non-English-centric training corpus
- gpt-4o-mini: closed-source, OpenAI instruction-tuning lineage; accessed via OpenAI API
- claude-haiku-4-5-20251001: closed-source, Anthropic instruction-tuning lineage

All models are prompted at temperature 0 for reproducibility using a system prompt and a structured user message:

System: You are a precise question-answering assistant. Answer the question using ONLY the information provided in the context. If the context does not contain enough information to answer the question, respond with exactly: I don’t know. Do not add information from your own knowledge. Keep your answer concise; one sentence or less.

User:

Context:

```
[1] {retrieved_context}
Question: {question}
Answer:
```

The prompt is identical for all six models, so observed differences in abstention and faithfulness cannot be attributed to prompt wording.

4.3 Metrics

Table 3 defines all six evaluation metrics.

Table 3: Evaluation metrics and desired direction on corrupted conditions. Formal definitions appear below.

Metric	Operationalization	Desired
Faithfulness	NLI entailment of the answer by retrieved context	Low
Exact match (EM)	Normalized match with the gold answer	Low
Token F1	Token-overlap F1 with the gold answer	Low
Hallucination	NLI-contradictory answer with low gold-answer overlap	Low
Abstention rate	Output starts with “I don’t know”	High
Contradiction rate	NLI contradiction between answer and retrieved context	High

For example i , let a_i denote a generated answer, c_i the retrieved context, g_i the gold answer, E_i the NLI entailment probability for (c_i, a_i) , and K_i the NLI contradiction indicator for (c_i, a_i) . Let M_i and F_i denote normalized exact match and token F1 against g_i , respectively. The reported condition-level metrics are means over the n examples. Faithfulness is

$$\text{Faithfulness} = \frac{1}{n} \sum_{i=1}^n E_i, \quad E_i = P_{\text{NLI}}(\text{entailment} \mid c_i, a_i). \quad (1)$$

The NLI model is cross-encoder/nli-deberta-v3-base. Exact match and token F1 are

$$\text{EM} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[M_i = 1], \quad \text{F1} = \frac{1}{n} \sum_{i=1}^n F_i. \quad (2)$$

An answer is counted as a hallucination only when it contradicts the retrieved context, has little overlap with the gold answer, and the answer itself contradicts the gold answer:

$$\text{Hallucination} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[K_i = 1 \wedge F_i < 0.2 \wedge \text{NLIContradict}(g_i, a_i) = 1]. \quad (3)$$

With $A_i = \mathbb{I}[a_i \text{ begins with “I don’t know”}]$, abstention and contradiction are

$$\text{Abstain} = \frac{1}{n} \sum_{i=1}^n A_i, \quad \text{Contradiction} = \frac{1}{n} \sum_{i=1}^n K_i. \quad (4)$$

The abstention-cliff score summarizes condition sensitivity:

$$\Delta_{\text{abstain}} = \frac{1}{2} [\text{Abstain}(\text{entity_swap}) + \text{Abstain}(\text{negation})] - \text{Abstain}(\text{original}) \quad (5)$$

A higher Δ_{abstain} indicates more abstention on corrupted than clean context. An ideal model would have a large positive score by abstaining on corrupted inputs while answering clean inputs.

The similarity-score ablation also reports an abstention-calibration discrepancy, adapted from expected calibration error. Partitioning outputs into ten bins b by E_i , with $\bar{A}_b$ and $\bar{E}_b$ denoting mean abstention and faithfulness in each bin, yields

$$\text{ECE}_{\text{abstain}} = \sum_b \frac{n_b}{n} |\bar{A}_b - (1 - \bar{E}_b)|. \quad (6)$$

This descriptive measure is distinct from the standard probability-calibration ECE because faithfulness, rather than a predicted probability, defines the target abstention rate.

4.4 Statistical Testing

All primary comparisons are paired because each question-answer pair appears in every condition. For a paired difference d_i , the paper reports the mean difference and a two-sided 95% confidence interval,

$$\bar{d} \pm 1.96 \frac{s_d}{\sqrt{n}}. \quad (7)$$

This interval is used for continuous faithfulness and binary-rate differences at $n = 1,000$. Abstention changes are additionally tested with exact two-sided McNemar tests. The 12 pre-specified original-versus-corrupted abstention tests are adjusted with the Holm procedure. Paraphrase comparisons are diagnostic and reported with confidence intervals rather than used to support the primary claims.

- Paired mean difference and 95% confidence interval for faithfulness, EM, F1, hallucination, and contradiction.
- Exact McNemar test for the primary binary abstention comparisons.
- Holm correction across the 12 primary abstention tests.

5 Results

5.1 Main Results

Table 4 presents the full results matrix across all six models and conditions ($N = 1,000$ per cell). Table 5 reports the paired confidence intervals and corrected tests for the primary comparisons.

5.2 RQ1: Do Models Follow Corrupted Context?

Faithfulness decreases from original to corrupted conditions across all six models. The paired 95% confidence intervals in Table 5 exclude zero in every primary comparison. llama3.1:8b has the highest baseline faithfulness (0.79) and decreases to 0.62 on entity swap and 0.55 on negation. mistral:7b starts lower (0.67) and falls most steeply to 0.40 on negation suggesting less tendency to follow context under contradiction. Among closed-source models, claude-haiku shows the lowest baseline EM (0.06) and F1 (0.32) on original context, indicating a more conservative answering style even on clean input.

Table 4: Main results by model and perturbation condition ($N = 1,000$ per cell). Values are condition means. Arrows indicate the desired direction on corrupted conditions. A value shown as $< .01$ denotes zero observed events; it does not imply a zero population rate.

Model	Condition	Faith $\downarrow$	EM $\downarrow$	F1 $\downarrow$	Abstain $\uparrow$	Halluc $\downarrow$	Contradict $\uparrow$
llama3.2:3b	Original	.74	.20	.47	.14	.04	.12
	Paraphrase	.71	.13	.37	.20	.05	.16
	Entity swap	.57	.12	.32	.30	.07	.25
	Negation	.51	.15	.35	.32	.04	.21
llama3.1:8b	Original	.79	.25	.55	.10	.03	.09
	Paraphrase	.74	.15	.38	.18	.04	.15
	Entity swap	.62	.16	.38	.28	.07	.23
	Negation	.55	.20	.41	.31	.04	.19
mistral:7b	Original	.67	.12	.41	.13	.03	.11
	Paraphrase	.63	.06	.30	.15	.03	.13
	Entity swap	.51	.07	.29	.26	.07	.24
	Negation	.40	.08	.26	.43	.04	.19
qwen2.5:7b	Original	.74	.14	.41	.16	.04	.12
	Paraphrase	.70	.12	.35	.19	.04	.14
	Entity swap	.53	.07	.26	.37	.09	.27
	Negation	.44	.10	.27	.43	.05	.24
gpt-4o-mini	Original	.74	.13	.43	.14	.03	.11
	Paraphrase	.69	.08	.32	.20	.05	.16
	Entity swap	.56	.09	.30	.32	.07	.26
	Negation	.45	.09	.27	.44	.05	.22
claude-haiku	Original	.72	.06	.32	.16	.01	.02
	Paraphrase	.67	.03	.23	.23	$< .01$	.03
	Entity swap	.54	.03	.22	.35	.02	.05
	Negation	.47	.02	.22	.46	$< .01$	.11

Paraphrase effects are smaller than corruption effects but are not negligible. Across models, the paired faithfulness change is -0.034 to -0.051 , compared with -0.153 to -0.295 for the corrupted conditions. EM also falls by 2.6–9.9 percentage points. This control therefore reveals surface-form sensitivity, and it should not be treated as an identical-to-original condition. The substantially larger effects of entity swap and negation remain the central result. Full diagnostics appear in Table 6.

5.3 RQ2: Do Models Hallucinate?

Hallucination rates are low across all six models and conditions, ranging from no observed events to 9%. Entity swap produces the highest observed rate for every model: qwen2.5:7b reaches 9%, and llama3.1:8b and mistral:7b each reach 7%. A plausible wrong entity may therefore be more likely to elicit confabulation than an explicit negation. Zero-event cells should be interpreted cautiously: with $n = 1,000$, the exact one-sided 95% upper confidence bound for a zero observed rate is 0.37%, rather than zero.

Table 5: Primary paired effects relative to original context ($n = 1,000$ per row). Faithfulness differences are mean differences with 95% confidence intervals; abstention differences are percentage points with paired 95% confidence intervals. All exact McNemar tests remain $q < .0001$ after Holm correction.

Model	Condition	Δ Faithfulness [95% CI]	Δ Abstention pp [95% CI]
llama3.2:3b	Entity swap	-0.172 [-0.198, -0.146]	+16.4 [+14.0, +18.8]
	Negation	-0.236 [-0.265, -0.207]	+18.3 [+15.7, +20.9]
llama3.1:8b	Entity swap	-0.173 [-0.198, -0.148]	+17.5 [+15.1, +19.9]
	Negation	-0.243 [-0.273, -0.213]	+20.9 [+18.2, +23.6]
mistral:7b	Entity swap	-0.153 [-0.179, -0.127]	+13.4 [+11.1, +15.7]
	Negation	-0.264 [-0.294, -0.234]	+30.9 [+28.0, +33.8]
qwen2.5:7b	Entity swap	-0.205 [-0.231, -0.178]	+20.2 [+17.6, +22.8]
	Negation	-0.292 [-0.322, -0.263]	+26.6 [+23.8, +29.4]
gpt-4o-mini	Entity swap	-0.180 [-0.204, -0.155]	+18.3 [+15.8, +20.8]
	Negation	-0.295 [-0.325, -0.265]	+30.0 [+27.1, +32.9]
claude-haiku	Entity swap	-0.174 [-0.199, -0.148]	+18.6 [+16.1, +21.1]
	Negation	-0.249 [-0.279, -0.219]	+30.2 [+27.3, +33.1]

Table 6: Paraphrase diagnostic: paired change relative to original context, with 95% confidence intervals ($n = 1,000$ per row). The smaller but consistently negative changes indicate surface-form sensitivity.

Model	Faithfulness Δ [95% CI]	EM Δ [95% CI]
llama3.2:3b	-0.034 [-0.057, -0.010]	-0.072 [-0.093, -0.051]
llama3.1:8b	-0.051 [-0.075, -0.027]	-0.099 [-0.123, -0.075]
mistral:7b	-0.039 [-0.065, -0.014]	-0.065 [-0.084, -0.046]
qwen2.5:7b	-0.035 [-0.056, -0.015]	-0.026 [-0.045, -0.007]
gpt-4o-mini	-0.050 [-0.071, -0.028]	-0.048 [-0.067, -0.029]
claude-haiku	-0.045 [-0.068, -0.022]	-0.026 [-0.038, -0.014]

claude-haiku has the lowest observed hallucination rates, from no observed events to 2%. The data do not establish the cause of this difference. Across all models, the larger and more consistent behavioral change is increased abstention rather than confabulation.

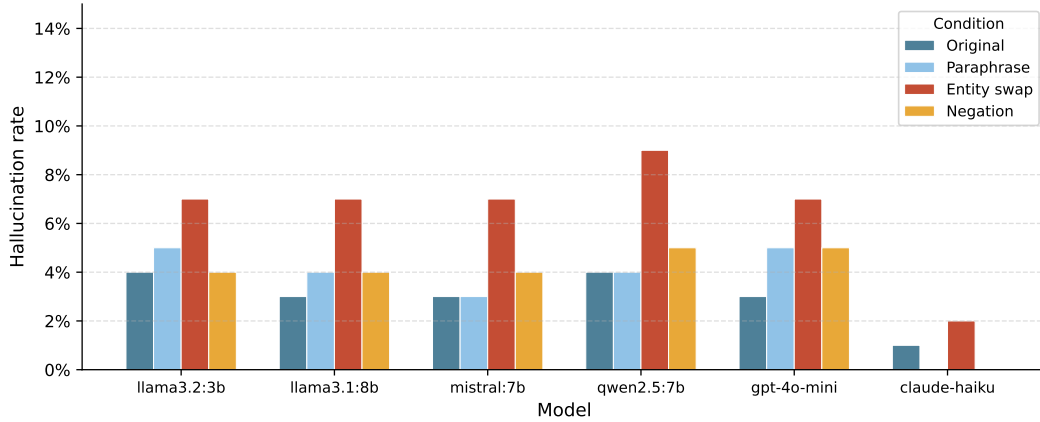


Figure 1: Hallucination rate by model and perturbation condition across all six models. Entity swap produces the highest observed hallucination rate consistently; claude-haiku has the lowest observed rates.

5.4 RQ3: Is Abstention Calibrated?

The abstention results are central to the benchmark. A well-calibrated model would show low abstention on original and paraphrase conditions and high abstention on corrupted conditions. Every primary paired comparison increases abstention, with 95% confidence intervals that exclude zero and Holm-adjusted $q < .0001$.

The Llama family shows the weakest calibration. llama3.2:3b abstains at only 14% on correct original context and reaches 30% on entity swap and 32% on negation ($\Delta_{\text{abstain}} = +0.173$). llama3.1:8b shows a similarly flat response (baseline 10%, rising to 28% and 31% on corrupted conditions, $\Delta_{\text{abstain}} = +0.192$). Both models commit readily on clean context but fail to substantially increase abstention under corruption.

Among open-weight models, mistral:7b and qwen2.5:7b show stronger calibration. mistral:7b rises from 13% baseline to 43% on negation, a 30 percentage-point increase, though entity swap abstention rises only modestly to 26%, reflecting the asymmetric failure mode described in Section 6.2. qwen2.5:7b shows a more balanced response: 37% on entity swap and 43% on negation ($\Delta_{\text{abstain}} = +0.234$).

Closed-source models achieve the strongest abstention cliffs. gpt-4o-mini rises from 14% baseline to 32% on entity swap and 44% on negation ($\Delta_{\text{abstain}} = +0.241$). claude-haiku rises from 16% to 35% and 46% respectively ($\Delta_{\text{abstain}} = +0.244$), the highest of any model evaluated. Both closed-source models show more balanced detection across entity swap and negation than their open-weight counterparts.

5.5 RQ4: Does Scale, Family, or Open- vs. Closed-Source Status Predict Calibration?

Table 7 reports cliff scores for all six models (see Equation (5) for the Δ_{abstain} definition).

Three observations emerge from this ranking. First, parameter count does not order calibration within this evaluated set: llama3.1:8b (8B parameters) scores lower than all three 7B competitors.

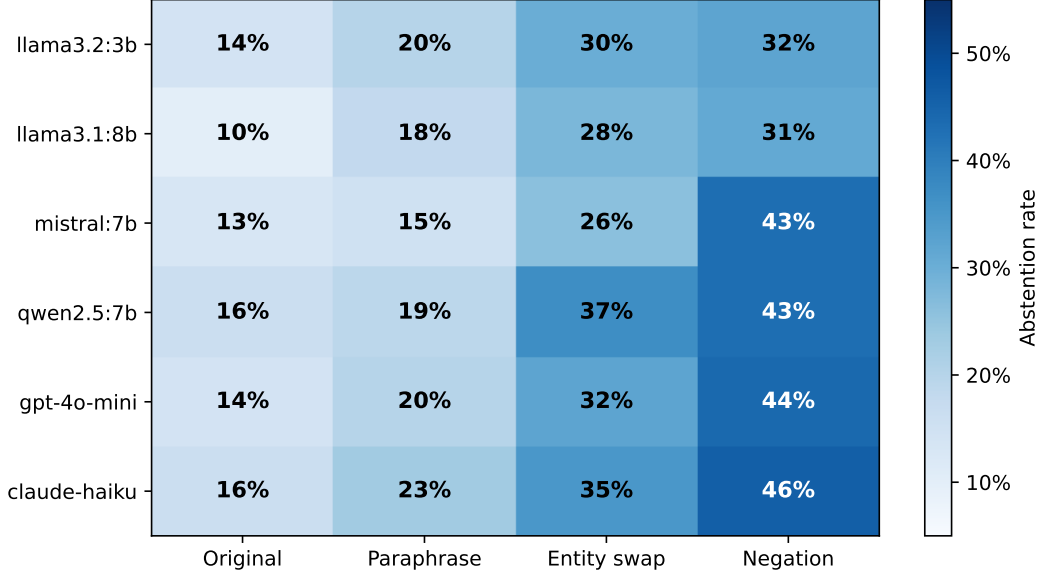


Figure 2: Abstention rates by model and condition across all six models. Closed-source models (gpt-4o-mini, claude-haiku) achieve the steepest cliffs; the Llama family exhibits the flattest response.

Table 7: Abstention-cliff scores by model, sorted by Δ_{abstain} . The score is descriptive; paired condition-level effects and confidence intervals are reported in Table 5.

Model	Type	Δ_{abstain}
llama3.2:3b	Open-weight	+0.173
llama3.1:8b	Open-weight	+0.192
mistral:7b	Open-weight	+0.222
qwen2.5:7b	Open-weight	+0.234
gpt-4o-mini	Closed-source	+0.241
claude-haiku	Closed-source	+0.244

Second, model family matters: the two Llama models cluster at the bottom regardless of size, while Mistral and Qwen score higher at a similar parameter count. Third, the two closed-source models occupy the top positions. Both gpt-4o-mini and claude-haiku outperform all four open-weight models on Δ_{abstain} despite the open-weight models spanning a comparable parameter range. This ranking is compatible with a role for model-family differences, but it does not identify a causal mechanism or establish a general open-versus-closed advantage.

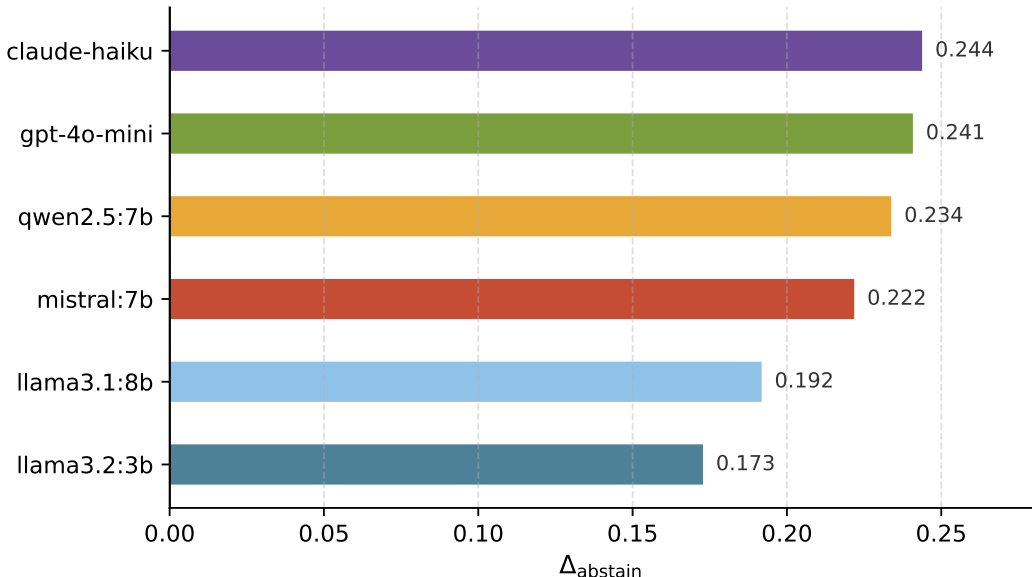


Figure 3: Abstention cliff score Δ_{abstain} by model. Closed-source models (gpt-4o-mini, claude-haiku) have the largest observed scores; llama3.2:3b has the smallest. Higher scores indicate more selective abstention across the evaluated conditions.

5.6 Retrieval Similarity Score as a Calibration Intervention

The results motivate an exploratory prompt-level intervention. The retriever similarity score is supplied as a prefix to the context field, allowing the model to condition its response on a retrieval-quality signal:

```
Context (with similarity prefix):
Retrieval confidence: 0.43 (low)
[1] {retrieved_context}
```

The intervention is evaluated on llama3.2:3b over a matched 200-example subset (800 model-condition inputs). Results are summarized in Table 8.

On the matched subset, the similarity prefix lowers the descriptive $\text{ECE}_{\text{abstain}}$ by 52% ($0.1080 \rightarrow 0.0516$). However, Δ_{abstain} decreases from $+0.165$ to $+0.135$. The prefix raises abstention on clean original context from 14.5% to 23.0%, almost as much as it raises abstention on corrupted conditions. It therefore makes the model more uniformly cautious rather than more condition-sensitive.

This single-model ablation suggests that the descriptive global discrepancy and selective calibration can move in different directions. Replication across models and a larger sample is needed before drawing a broader conclusion.

Table 8: Matched similarity-score ablation on llama3.2:3b ($n = 200$ question-answer pairs; 800 model-condition inputs). ECE_{abstain} is defined in Equation (6); Δ_{abstain} is defined in Equation (5). This single-model result is descriptive.

Condition	ECE	Δ_{abstain}	Orig. Abstain
Baseline (no similarity)	0.1080	+0.165	0.145
With similarity score	0.0516	+0.135	0.230

6 Analysis

6.1 Failure Mode Taxonomy

The joint distribution of faithfulness, EM, hallucination, and abstention motivates three descriptive generator archetypes:

- Context-follower. High faithfulness on corrupted conditions, low EM. The model accepts the retrieved context even when it is wrong and produces answers consistent with the incorrect passage. Errors of this type are difficult to detect in production because the outputs are coherent and internally consistent. No evaluated model is a pure context-follower, but all show partial context-following on entity swap (faithfulness 0.51–0.62 across open-weight models).
- Hallucinator. Low faithfulness, low EM, elevated hallucination rate. The model disregards the retrieved context and produces a plausible-sounding answer from parametric memory. This behavior is uncommon in the observed data: hallucination peaks at 9%. claude-haiku has the lowest observed rate, from no observed events to 2% across conditions.
- Over-abstainer. High uniform abstention regardless of context quality. The model declines to answer at similar rates on clean and corrupted context alike. All six models show some degree of over-abstainer behavior. The Llama family shows it most clearly: llama3.2:3b abstains at 14% on correct original context and produces a cliff score of only $\Delta_{\text{abstain}} = +0.173$, the weakest of any model evaluated.

An ideal model would commit on clean context and abstain on corrupted context, producing a large positive cliff score. The observed scores range from +0.173 to +0.244. All six models therefore show room for more selective abstention, but this sample does not support a claim about all current LLM families.

6.2 Mistral’s Asymmetric Response to Failure Modes

mistral:7b shows an asymmetric response across corruption types. It abstains at only 26% on entity swap (barely above its 13% baseline) but rises to 43% on negation, a 30 percentage-point increase. Mistral is particularly sensitive to explicit logical contradiction but is less responsive to entity-level substitutions. Direct factual contradictions therefore produce a larger abstention change than plausible wrong entities in this model.

qwen2.5:7b shows a more balanced response: 37% on entity swap and 43% on negation. Its observed cliff score is the largest among the four open-weight models, although both closed-source models have higher scores in this six-model comparison.

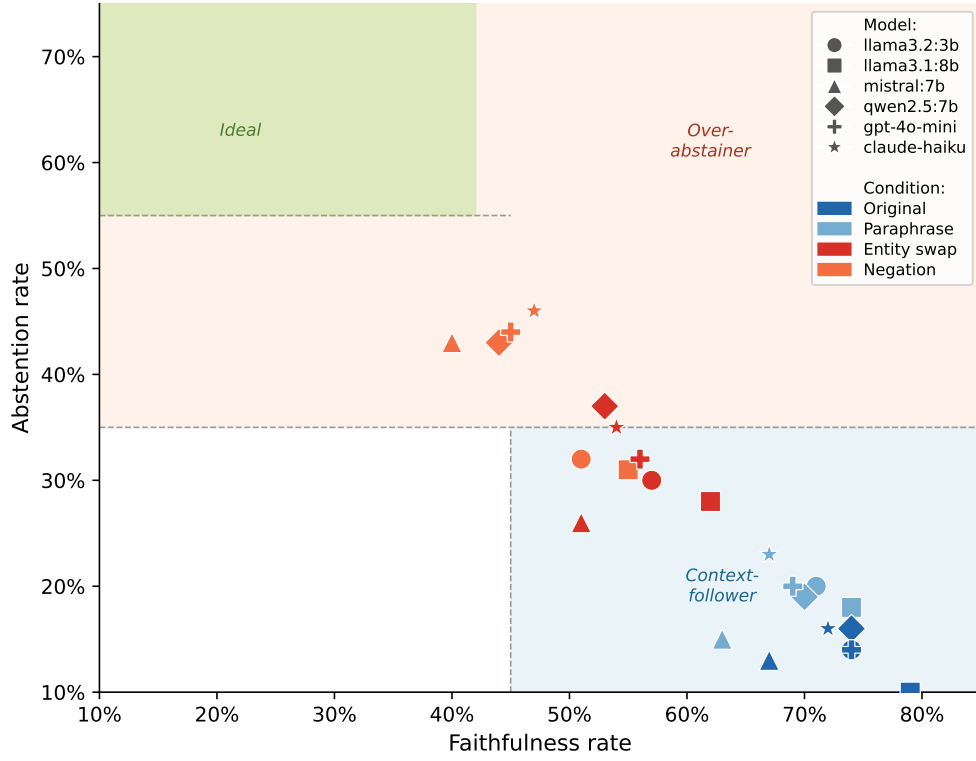


Figure 4: Failure mode taxonomy. Each point is a (model, condition) pair plotted by faithfulness rate (x) and abstention rate (y) across all six models. Shaded regions mark the three archetypes. Corrupted conditions (entity swap, negation) move left and up relative to original/paraphrase, but many remain in the over-abstainer region rather than the ideal (low faithfulness, high abstention) region.

6.3 Answer-Type Cross-Tabulation

The expanded dataset enables analysis of failure mode distributions across answer types. Drawing on the error analysis output across all six models, three patterns emerge.

Organization-type answers show the highest hallucination rates on entity swap across models. This is consistent with the hypothesis that organization names are well-represented in parametric memory: when a retrieved org name is wrong, models are more likely to substitute a known alternative from training data rather than abstaining or following the corrupted context.

Location-type answers show the lowest abstention rates on entity swap despite corruption. Models therefore more often accept and echo a plausible wrong location than abstain. A wrong city or country name is structurally indistinguishable from a correct one without world knowledge, which may make location swaps difficult to detect from context alone.

Date and number answers show the highest context-following rates on negation. When a negated sentence still contains a specific number (e.g., “did not send 126 guns”), models frequently extract the number despite the negation, producing answers that are faithful to the surface form but semantically incorrect. This represents a systematic blind spot: the presence of a numeric anchor overrides sensitivity to the negation operator.

6.4 Paraphrase as a Diagnostic

The paraphrase diagnostic confirms that surface wording affects outputs. The faithfulness change ranges from -0.034 to -0.051 across models, and the EM change ranges from -0.026 to -0.099 . These values are smaller than those for entity swap and negation, but they show that paraphrase is a control with measurable variation rather than a perfectly invariant condition.

6.5 Error Analysis

Entity-swap and negation failures are examined across all six models at $N = 1,000$ (see Appendix B for full category counts). Errors fall into three categories: (a) the model answered correctly from parametric memory despite corrupted context, (b) the model accepted the corrupted context and produced a wrong answer, or (c) the model abstained.

Category (a): Correct answer despite corrupted context. The model outputs the gold answer even though the corrupted context does not support it. This category can reflect parametric memory or an incomplete perturbation and therefore is not treated as direct evidence of calibrated behavior.

Category (b): Followed corrupted context. The model produces an answer entailed by the wrong context and therefore incorrect. This is the most actionable failure mode: the output is coherent and high-confidence, making errors difficult to detect downstream.

Category (b): Context-follower on entity swap

Question: What threat does the Global Adaptation Index study?

Perturbed context (entity swap): As of 2013, the university is home to the Notre Dame Global Adaptation Index which ranks countries 2010 based on how vulnerable they are to climate change and how prepared they are to adapt.

Model output (qwen2.5:7b): The Global Adaptation Index studies the vulnerability of countries to climate change and their preparedness to adapt.

Faithfulness: 0.99 EM: false Perturbation: entity swap

The swapped entity left the claim structure intact; the model faithfully echoed the corrupted context without detecting the substitution.

Category (c): Appropriate abstention on corrupted context.

Category (c): Correct abstention on negation

Question: In what year did Jerome Green send his first wireless message?

Perturbed context (negation): Around 1899, Professor Jerome Green did not become the first American to send a wireless message.

Model output (qwen2.5:7b): I don't know.

Faithfulness: 0.05 EM: false Perturbation: negation

The explicit logical contradiction triggered abstention; the model correctly declined to commit to an answer.

7 Discussion

Over-abstention as miscalibration. All six models abstain at non-trivial rates on clean original context (10–16%). Given the prompt’s instruction to abstain only when context is insufficient, these responses represent lost coverage on answerable inputs. Well-calibrated behavior requires low clean context abstention together with higher abstention on corrupted context, rather than a uniformly cautious response.

Closed-source models show stronger calibration, but the gap is narrow. The two closed-source models have the highest observed cliff scores, 0.241 and 0.244, while open-weight scores range from 0.173 to 0.234. The margin is narrow: the difference between the best open-weight and best closed-source model is 0.010. This six-model sample does not establish a general closed-source advantage.

Model family over scale. llama3.1:8b (8B parameters) does not exceed its 7B competitors on the abstention cliff despite achieving the highest baseline faithfulness (0.79). Within this small model set, parameter count alone does not order abstention calibration. The data are insufficient to identify whether model family, training data, alignment method, or another design choice accounts for the ranking.

Implications for RAG system design. Model selection for RAG should assess clean-context coverage and response to corrupted context together. The abstention cliff complements standard answer-quality metrics because it captures their difference across retrieval conditions. The similarity-score result in Section 5.6 further shows that a lower descriptive calibration discrepancy does not necessarily imply improved selective abstention.

8 Limitations

- Language. All perturbations, models, and NLI scorers are English-only. Abstention calibration behavior may differ substantially across languages, particularly for models with non-English-centric training corpora such as qwen2.5:7b.

- Entity swap coverage. The lookup table covers common named entity types; rare or domain-specific entities (medical, legal, technical) may not have suitable replacements. Human validation confirmed that entity swap failures are concentrated in chemical compound names and other domain-specific terms outside the lookup table’s scope.
- Negation coverage. The dependency-parse negation handles simple declarative sentences; complex, compound, or passive constructions may produce less natural negations, which remain in the evaluation set. Human validation identified multi-clause sentences as a systematic source of negation failures, where negating the root verb creates internal logical contradictions with dependent clauses.
- Paraphrase quality. The LLM paraphraser does not reliably preserve quoted speech verbatim, altering words attributed to named individuals. This is a known failure mode of instruction-tuned paraphraser and is reflected in the human validation correctness scores.
- Closed-source model opacity. gpt-4o-mini and claude-haiku are accessed through provider interfaces, so their training data, instruction-tuning methodology, or RLHF signal. The closed-source advantage in abstention calibration is an empirical observation; causal attribution to specific training factors is not possible without access to model internals.
- Similarity score experiment scope. The retrieval similarity score ablation is conducted on a 200-example subset of one model (llama3.2:3b). Whether the dissociation between global calibration (ECE) and selective calibration (Δ_{abstain}) generalizes to other models or larger subsets remains an open question.
- Single dataset. All experiments use SQuAD v1.1, which consists of Wikipedia passages with extractive answers. Abstention behavior may differ on open-ended, multi-hop, or domain-specific QA tasks where the answer is not directly extractable from a single context sentence.
- Single retriever. All experiments use FAISS with all-MiniLM-L6-v2. Different retriever architectures or similarity functions may change retrieved passage distributions and affect results.
- NLI scorer assumptions. The DeBERTa-v3 NLI model is imperfectly calibrated for short factoid answers and may misclassify entailment in edge cases, particularly for numeric answers where a negated sentence still contains the correct number as a surface-form anchor.

9 Conclusion

This paper presents a counterfactual benchmark for measuring how RAG generators respond to controlled retrieval failure. By holding the question, answer, and prompt fixed while varying context content across original, paraphrase, entity-swap, and negation conditions, the design isolates generator response to retrieval content across six models and 1,000 paired examples.

Faithfulness decreases by 0.153–0.295 on corrupted contexts, and abstention increases by 13.4–30.9 percentage points; every primary paired 95% confidence interval excludes zero. Hallucination is less frequent, with observed rates from no events to 9%. The two closed-source models have the largest observed abstention cliffs, while the Llama models have the smallest. The matched similarity-score ablation reduces a descriptive calibration discrepancy but also reduces the selective-abstention score. These findings identify selective abstention as a measurable weakness under controlled retrieval failure.

Future work. Future evaluation should test additional QA datasets, multilingual contexts, more model families, and larger matched similarity-score studies. Controlled training interventions for selective abstention also require direct evaluation before claims about their efficacy are warranted.

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A Prompt Templates

Generator prompt.

System: You are a precise question-answering assistant. Answer the question using ONLY the information provided in the context. If the context does not contain enough information to answer the question, respond with exactly: I don't know. Do not add information from your own knowledge. Keep your answer concise; one sentence or less.

User:

Context:

[1] {retrieved_context}

Question: {question}

Answer:

Paraphrase generation prompt.

System: You are a precise paraphrasing assistant. Rewrite the given sentence using different wording, but preserve every factual detail exactly: names, dates, numbers, and all other specifics must remain correct. Return only the rewritten sentence with no extra commentary.

User: {gold_context}

Similarity score prompt (ablation only).

System: You are a precise question-answering assistant. Answer the question using ONLY the information provided in the context. If the context does not contain enough information to answer the question, respond with exactly: I don't know. Do not add information from your own knowledge. Keep your answer concise; one sentence or less.

User:

Retrieval confidence: {similarity_score} (low | medium | high)

Context:

[1] {retrieved_context}

Question: {question}

Answer:

B Error Analysis Breakdown

Table 9 reports the full error-analysis category counts across all six models and both corrupted perturbation types ($N = 1,000$ per cell, 6 models = 6,000 records per perturbation type).

Category (b) is the most actionable failure mode. Context-following on entity swap (2,780 instances, 46.3%) substantially exceeds context-following on negation (2,085 instances, 34.8%), consistent with the finding that plausible wrong entities are harder to detect than explicit logical contradictions. When a retrieved document contains a structurally correct but factually wrong entity, models largely accept and echo the corrupted content with high faithfulness and no signal of uncertainty.

Appropriate abstention (category c) is more common on negation (40.0%) than entity swap (31.2%), reflecting the same asymmetry observed in the cliff score analysis: explicit logical contradiction is a stronger abstention trigger than entity-level substitution. The concern raised in

Table 9: Error analysis category counts (all six models combined, corrupted conditions only). Cat. (a): correct despite corrupted context (parametric memory, EM true). Cat. (b): followed corrupted context (context-follower, faithfulness > 0.70, EM false). Cat. (c): abstained on corrupted condition. “Other”: wrong answer, low faithfulness, not abstained.

Category	Entity swap	%	Negation	%
(a) Parametric memory (correct)	546	9.1%	627	10.5%
(b) Context-follower (wrong)	2,780	46.3%	2,085	34.8%
(c) Appropriate abstention	1,874	31.2%	2,399	40.0%
Other (wrong, low faith)	800	13.3%	889	14.8%
Total	6,000	100%	6,000	100%

Section 5.4 is that these same models also abstain at non-trivial rates on clean original context (10–16%), meaning the category (c) counts include both well-calibrated abstentions on corrupted input and over-abstentions that would also occur on clean input.

Parametric memory override (category a) is more common on negation (10.5%) than entity swap (9.1%). This is consistent with the hypothesis that negated sentences still contain the correct factual anchor, such as a number, name, or date, which models retrieve from parametric memory despite the negation operator, producing a technically correct answer through the wrong mechanism.